Meeting Notes & System Architecture (Oct 12)

1. Core Objective: High-throughput Handwritten OCR pipeline.

2. Pipeline requirements:

   - Multi-page ingest (PDF, multi-image scan, mobile photos).

   - Shadow removal and deskewing using adaptive filters.

   - Parallel batch workers for high-volume production.

3. Benchmarks & accuracy metrics:

   - Cursive text recognition target > 95% accuracy.

   - Latency: under 2.5 seconds per page with parallel scaling.

~~Old approach: basic Tesseract 4~~ -> Discarded due to poor cursive recall.

Action item: Hook up Gemini Vision API + EasyOCR local fallback.

Mathematical Formulas & Derivations

Loss Function Definition:

L_total = lambda_1 * L_ctc + lambda_2 * L_seq2seq + lambda_3 * L_attention

Key parameters for deskewing:

theta = median(arctan(-dy / dx)) for all detected line contours.

Range: -45 deg <= theta <= 45 deg.

Illumination Correction:

I_norm(x, y) = 255 - |I(x, y) - Blur(Dilate(I(x, y), K))|

Verified: Contrast ratio boosts ink legibility by ~38%.

Production Deployment Checklist

[X] Asynchronous queue management with concurrency limiter.

[X] PyMuPDF 300 DPI high-definition page extraction.

[X] Human-in-the-loop review studio with side-by-side verification.

[X] Export formats: Markdown, Plain Text, Structured JSON, ZIP bundle.

[ ] Final stress testing with 50+ page continuous notebook scans.

Sign-off: ParallaxFlow Engineering Team